

WONHO BAE

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My research focuses on **data-centric and data-efficient machine learning**, with the goal of improving learning when data is limited, noisy, or expensive to obtain. I develop principled methods for **measuring data importance**, **evaluating data quality**, and **improving learning efficiency** through better use of data. More broadly, my work spans learning under limited supervision, including **active learning**, **weakly supervised learning**, and **meta-learning**, and explores both **discriminative and generative models**, e.g., LLM, grounded in statistical learning and deep learning theory. Current interests include data selection, personalization and federated learning.

RESEARCH EXPERIENCE

Apple ASR Data-Centric ML Team

August 2025 - Present

AI/ML Resident

- Supervisor: Dr. Honza (Jan) Silovsky
- Conducted researches on semi-supervised federated learning to improve data efficiency, differential privacy and personalization for automated speech recognition tasks.

Machine Learning Lab at University of British Columbia

September 2020 - August 2025

Research Assistant

- Supervisor: Prof. Danica J. Sutherland
- Conducted researches on learning under low supervision such as i) semi and weakly-supervised learning, ii) meta learning, iii) active learning, as well as iv) deep learning theory. Recent focus is on data efficient learning for in-context and preference learning in Large Language Models (LLMs).

Borealis AI at Vancouver

May 2022 - September 2022, May 2023 - November 2024

Research Intern

- Supervisor: Dr. Gabriel Oliveira, Dr. Fred Tung, and Dr. Mohamed Ahmed
- Conducted researches on i) temporal point processes (TPPs) to capture periodic patterns using neural process framework, ii) TPPs to predict long-term horizon using group-wise diffusion models, iii) difficulty-based regularization method, and iv) active learning robust to various budget regimes

Vision & Learning Lab at Seoul National University

February 2018 - September 2020

Research Assistant

- Supervisor: Prof. Gunhee Kim
- Conducted researches on i) small object detection using GANs in the Faster R-CNN framework, and ii) object localization task under weakly-supervised learning setting using a class activation mapping method

Data Science for Common Good Fellowship at UMass, Amherst

May 2019 - August 2019

Research Fellow

- Supervisor: Dr. Brant Cheikes, Prof. Matthew Rattigan
- Conducted a research on classifying wild animal images collected using camera traps in collaboration with The Nature Conservancy. Deployed a web-based open-source tool for ecologists

Renewable & Appropriate Energy Lab at UC Berkeley

January 2017 - December 2017

Research Assistant

- Supervisor: Prof. Daniel Kammen, Prof. Deborah Sunter
- Participated in the Inclusive Green Growth project. Worked on keyword detection task using Natural Language Process techniques to replace synonyms and pronouns in the text

EDUCATION

University of British Columbia

September 2020 - November 2025

PhD of Computer Science

GPA: 4.00

University of Massachusetts, Amherst
Master of Computer Science

September 2018 - May 2020

GPA: 3.78

University of California, Berkeley
Bachelor of Statistics

September 2013 - December 2017

GPA: 3.76

Santa Monica College
Associate of Economics, member of Alpha Gamma Sigma

September 2011 - May 2013

GPA: 3.95

PUBLICATIONS

(* denotes equal contribution and ** denotes an important note)

- [1] Giyeong Oh, Junghyun Lee, Jaehyun Park, Youngjae Yu, Junhyug Noh*, **Wonho Bae***, “Random Is Hard to Beat: Active Selection in Online DPO with Modern LLMs”, **ICLRW 2026**, Rio de Janeiro, April 2026.
- [2] Jeongin Kim, **Wonho Bae**, YouLee Han, Giyeong Oh, Youngjae Yu, Danica J. Sutherland, Junhyug Noh, “Diffusion-Driven Two-Stage Active Learning for Low-Budget Semantic Segmentation”, **NeurIPS 2025**, San Diego, December 2025.
- [3] Jing Wang, **Wonho Bae**, Jiahong Chen, Wenxu Wang, Junhyug Noh, “Vicinity-Guided Discriminative Latent Diffusion for Privacy-Preserving Domain Adaptation”, **NeurIPS 2025**, San Diego, December 2025.
- [4] Jihyun Yu, Yoojin Oh, **Wonho Bae**, Mingyu Kim, Junhyug Noh, “SteeringTTA: Guiding Diffusion Trajectories for Robust Test-Time-Adaptation”, **ICMLW 2025**, Vancouver, July 2025.
- [5] **Wonho Bae**, Gabriel L. Oliveira*, Danica J. Sutherland*, “Uncertainty Herding: One Active Learning Method for All Label Budgets”, **ICLR 2025**, Singapore, April 2025.
Note: This work has been filed for **U.S. patent, November 2024.
- [6] **Wonho Bae**, Junhyug Noh, Danica J. Sutherland, “Generalized Coverage for More Robust Low-Budget Active Learning”, **ECCV 2024**, Milano, October 2024.
- [7] **Wonho Bae**, Jing Wang, Danica J. Sutherland, “Exploring Active Learning in Meta-Learning: Enhancing Context Set Labeling”, **ECCV 2024**, Milano, October 2024.
- [8] **Wonho Bae**, Yi Ren, Mohamed Osama Ahmed, Frederick Tung, Danica J. Sutherland, Gabriel L. Oliveira, “AdaFlood: Adaptive Flood Regularization”, **TMLR 2024**.
- [9] Jing Wang, **Wonho Bae**, Jiahong Chen, Kuangen Zhang, Leonid Sigal, “What Has Been Overlooked in Contrastive Source-Free Domain Adaptation: Leveraging Source-Informed Latent Augmentation within Neighborhood Context”, **TMLR 2024**.
- [10] Mohamad Amin Mohamadi, **Wonho Bae**, Danica Sutherland, “A Fast, Well-Founded Approximation to the Empirical Neural Tangent Kernel”, **ICML 2023**, Hawaii, July 2023.
- [11] **Wonho Bae**, Mohamed Osama Ahmed, Frederick Tung, Gabriel Leivas Oliveira, “Meta Temporal Point Processes”, **ICLR 2023**, Kigali Rwanda, May 2023.
Note: This work has been further developed into a **U.S. patent, March 2025.
- [12] Yi Ren, Shangmin Guo, **Wonho Bae**, Danica J. Sutherland, “How to Prepare Your Task Head for Finetuning”, **ICLR 2023**, Kigali Rwanda, May 2023.
- [13] Junhyug Noh, Kyung Don Yoo, **Wonho Bae**, ..., YonSu Kim, Gunhee Kim, “Predicting outcomes of continuous renal replacement therapy using body composition monitoring: a deep-learning approach”, **Scientific Reports (2023)** by Nature Publishing Group.
- [14] **Wonho Bae***, Mohamad Amin Mohamadi*, Danica Sutherland, “Making Look-Ahead Active Learning Strategies Feasible with Neural Tangent Kernels”, **NeurIPS 2022**, New Orleans, November 2022.
- [15] Jinhwan Seo, **Wonho Bae**, Danica J. Sutherland, Junhyug Noh, Daijin Kim “Object Discovery via Contrastive Learning for Weakly Supervised Object Detection”, **ECCV 2022**, Tel-Aviv, October 2022.

- [16] **Wonho Bae**, Junhyug Noh, Milad Jalali Asadabadi, Danica J. Sutherland, “One Weird Trick to Improve Your Semi-Weakly Supervised Semantic Segmentation Model”, **IJCAI 2022**, Vienna, July 2022.
- [17] **Wonho Bae***, Junhyug Noh*, Gunhee Kim, “Rethinking Class Activations Mapping for Weakly Supervised Object Localization”, **ECCV 2020**, online, August 2020.
- [18] Junhyug Noh, Kyung Don Yoo, **Wonho Bae**, ..., YonSu Kim, Gunhee Kim, “Prediction of the Mortality Risk in Peritoneal Dialysis Patients using Machine Learning Models: A Nation-wide Prospective Cohort in Korea”, **Scientific Reports (2020)** by Nature Publishing Group.
- [19] Junhyug Noh, **Wonho Bae**, Wonhee Lee, Jinhwan Seo, Gunhee Kim, “Better to Follow, Follow to Be Better: Towards Precise Supervision of Feature Super-Resolution for Small Object Detection”, **ICCV 2019**, Seoul, October 2019.

WORK EXPERIENCE

OmniSync AI

October 2023 - June 2025

Co-founder & Research Lead

- Founded AI startup, providing lip-synchronization service. Has led research and development to develop core AI models for lip-sync based on state-of-the-art generative models.

Republic of Korea Army

February 2015 - November 2016

Signals Intelligence Analyst

- Served in the intelligence battalion of the Republic of Korea Army for 21 months as a signals intelligence analyst.

AWARD & SCHOLARSHIP

Mitacs program funding for two terms – CAD 30K

March - November 2024

Mitacs funded research partnering with Borealis AI

NSERC AML-TN graduate trainee support – CAD 32K

October 2024 - October 2025

Graduate trainee program with 1 year full stipend support

Learning from Imperfect Data (LID) Competition – 1st

June 2020

1st place in LID workshop at CVPR 2020

Data Science for Common Good Fellowship

May 2019 - August 2019

Research fellow in the Center of Data Science at UMass, Amherst

American Math Competitions

2011 - 2012

3rd place in 2011 and 1st place in 2012

TEACHING

Teaching Assistant

Advanced Machine Learning (UBC: 2023), Computer Vision (UMass: 2019, UBC: 2021, 2022)

OUTREACH / PRESENTATIONS

Talk at Vancouver Machine Learning Reading Group at UBC

March 2025

Presented multiple works on active learning mainly focusing on MaxHerding and UHerding.

Talk at Borealis AI, Vancouver

September 2023

Presented the work on “Meta Temporal Point Processes”, ICLR 2023 at Borealis-UBC workshop.

Talk at ViewMagine (Online)

January 2021

Presented “How to Access a Research Problem in Computer Vision” based on the publications from ICCV 2019 and ECCV 2020 and research design course in UMass.

AI Summer Seminar at UMass, Amherst*Summer 2019*

Hosted AI seminar at UMass during Summer of 2019. Discussed various topics related to AI including but not limited to computer vision, natural language process and planning.

Presentation for Inclusive Green Growth at Institute of Advanced Study, Germany*August 2018*

Presented a data-driven approach for measuring Inclusive Green Growth of different countries and regions at Hanse-Wissenschaftskolleg Institute for Advanced Study in Germany.

SERVICES

Paper Review

NeurIPS (2021-2025), ICML (2022-2026), ICLR (2023-2026), CVPR (2022-2023), TMLR (2025), AAAI (2025)

Recruiting Committee

UBC Graduate Recruiting and Admissions Committee (2023)

European Laboratory for Learning and Intelligent Systems (ELLIS) PhD Committee (2024)

Volunteer*September 2022 - December 2022*

Mentor in Science Undergraduate Society Mentorship Program at UBC